natural system. Although its location has meant that detailed studies of the area are limited, the Coral Sea is a known habitat for many protected species, and spawning aggregations and nesting locations have been identified. The area also provides migratory corridors for a variety of important species, and as such provides suitable baseline conditions for future monitoring studies.

RESOLUTION MEPC.268(68)
(adopted on 15 May 2015)
DESIGNATION OF THE SOUTH-WEST CORAL SEA AS AN EXTENSION OF THE GREAT BARRIER REEF AND TORRES STRAIT PARTICULARLY SENSITIVE SEA AREAMEPC 68/21/Add.1
Annex 14, page 11

https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
ANNEX 3

VULNERABILITY TO DAMAGE BY INTERNATIONAL SHIPPING ACTIVITIES2

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SOURCE: MEPC.268(68).pdf
TORRES STRAIT PARTICULARLY SENSITIVE SEA AREAMEPC 68/21/Add.1
Annex 14, page 11

https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
ANNEX 3

VULNERABILITY TO DAMAGE BY INTERNATIONAL SHIPPING ACTIVITIES2

(Note: A detailed description of the characteristics of the maritime traffic, the transport of harmful substances, and the threats from maritime incidents, including a description of the hydrographical, meteorological and oceanographical condit ions may be found in paragraphs 81 to 109 of document MEPC 68/10/1.)

1 Vessel traffic characteristics

Operational factors

1.1 There are two major shipping routes in the region the Inner Route and the Outer Route of the GBR. While the Inner Route lies relatively close to the Queensland coast within the GBR and Torres Strait PSSA, the Outer Route begins at the north-eastern limit of the Torres Strait (the Great North-East Channel), continues southwards through the Coral Sea and re-joins the Queensland coast near Sandy Cape (south of Gladstone) (see figure 1 of appendix 3 of document MEPC 68/10/1).

1.2 The Outer Route experiences south-east trade winds and heavy seas for about nine

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SOURCE: MEPC.268(68).pdf
and re-joins the Queensland coast near Sandy Cape (south of Gladstone) (see figure 1 of

appendix 3 of document MEPC 68/10/1).

1.2 The Outer Route experiences south-east trade winds and heavy seas for about nine months of the year. A vessel suffering serious propulsion or power failure in the Coral Sea will be many hundreds of kilometres from towage assistance and could drift on to one of the numerous reefs or cays in the Coral Sea before any towage assistance can arrive. Anchoring is impractical due to the precipitous depths that prevail up to the edges of these reefs.

Vessel types

1.3 There is a wide variety of vessel types operating in this area. Ships entering and leaving Queensland coastal ports are primarily dry bulk carriers (most notably carrying coal) and, increasingly, liquefied natural gas (LNG) tankers. Oil and chemical carriers calling at Australian east coast ports mainly choose to use the Outer Route. Other ships transiting through the south west Coral Sea trading between Asia and other east coast Australian ports, such as Brisbane, Newcastle, Sydney and Melbourne, transport a variety of cargoes

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SOURCE: MEPC.268(68).pdf
through the south west Coral Sea trading between Asia and other east coast Australian ports, such as Brisbane, Newcastle, Sydney and Melbourne, transport a variety of cargoes including containerized, dry, liquid, vehicular and general cargoes.

Traffic characteristics

1.4 In addition to ships using the Outer Route to transit between Torres Strait and east coast Australian ports, the Outer Route converges in the south western Coral Sea with the north/south route used by ships transiting between Asian ports via Jomard Entrance (Papua New Guinea) and major Australian east coast ports such as Newcastle, Sydney and Melbourne, and commodity exporting ports in Queensland.

1.5 Ships bound to and from Queensland ports are also a major consideration in terms of the risk to the ecosystem from international shipping activities in the south west Coral Sea. There are four main passages through the GBR that result in a corresponding concentration of traffic in the south west Coral Sea. These are Grafton Passage (near Cairns); Palm Passage (near Townsville); Hydrographers Passage (near Mackay); and through the Capricorn and Curtis Channels in the south.

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SOURCE: MEPC.268(68).pdf
of traffic in the south west Coral Sea. These are Grafton Passage (near Cairns); Palm Passage (near Townsville); Hydrographers Passage (near Mackay); and through the Capricorn and Curtis Channels in the south.

2 The text in this annex is taken from Australia's submission contained in document MEPC 68/10/1.

https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
1.6 These shipping routes and passages can be identified in the Automatic Identification System (AIS) vessel traffic density information provided in appendix 2 of document MEPC 68/10/1.

1.7 Shipping activity in the Coral Sea is expected to increase in the coming years. The expansion of the Australian resources sector, which includes other east coast bulk ports such as Newcastle and Port Kembla (most ships en route to and from these ports use the

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SOURCE: MEPC.268(68).pdf
expansion of the Australian resources sector, which includes other east coast bulk ports such as Newcastle and Port Kembla (most ships en route to and from these ports use the north/south route through the south west Coral Sea), is the major factor in the expected growth of 81% in the total national traffic at sea by 2020.

Harmful substances carried

1.8 A wide variety of vessels carrying a range of potentially harmful substances operate in this area. The Outer Route is generally used by oil and chemical tankers visiting Australian east coast ports, while there is increasing LNG tanker traffic entering and leaving Queensland ports.

2 Natural factors

Hydrographical

2.1 There are some areas immediately around Coral Sea cays, reefs and islets where the depth of water, surveyed bathymetry quality and/or final charted product scale pose some navigational risk for larger vessel types. These areas are clearly marked on nautical charts and, in general, such areas should be well avoided by commercial shipping. Shipping should always navigate with due regard for charted data, chart scale and stated reliability of data within the area of interest.

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SOURCE: MEPC.268(68).pdf
charts and, in general, such areas should be well avoided by commercial shipping. Shipping should always navigate with due regard for charted data, chart scale and stated reliability of data within the area of interest.

2.2 Generally, all areas within the PSSA (apart from the proposed area to be avoided (ATBA), discussed below) are too deep to offer any anchoring opportunity.

Meteorological

2.3 Parts of the area are subject to the highest frequency of tropical cyclones in eastern Australia, creating a high disturbance regime for its ecological communities, resulting in increased diversity due to the frequent regeneration of reefs following storm events. In the last 100 years, cyclones have become less frequent but more intense, and in the last 12 years the region has experienced four extreme cyclones (category 4 or 5).

2.4 In general, during winter months, the predominant winds are from the south-east with small southerly and easterly components. Summer months exhibit an increased easterly component in addition to the south easterly winds.

2.5 Winds in these areas may produce shallow surface currents in addition to those

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SOURCE: MEPC.268(68).pdf
with small southerly and easterly components. Summer months exhibit an increased easterly component in addition to the south easterly winds.

2.5 Winds in these areas may produce shallow surface currents in addition to those deeper currents described below.

Oceanographic

2.6 Oceanographic processes play a significant role in the biological patterns across the Coral Sea region. There are three main currents that affect the region; the South Equatorial, Hiri, and East Australian currents. The South Equatorial Current moves west toward Townsville and Cairns from offshore waters to the east. As it moves toward the coast it splits into the north-flowing Hiri Current and the East Australian Current. Geomorphic features interact with these ocean currents to create variable speeds and directions.